

Atharv Shah

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GitHub | LinkedIn

EXPERIENCE

- **AI Research Intern — Breast Cancer Computer-Aided Diagnosis** LNMIIT
Machine learning and computer vision workflows for breast cancer analysis
 - Developed reproducible Python workflows spanning image preprocessing, feature engineering, model training, quantitative evaluation, and error analysis across medical imaging datasets.
 - Investigated data/model quality and failure patterns to improve reliability and support systematic validation.
- **Software Intern — Industrial Automation**
Python, SQL, data processing, transformation, and validation
 - Developed Python/SQL workflows integrating SQL Server data and automating extraction, cleaning, transformation, and validation for downstream analysis.
 - Applied version-control and software engineering practices to deliver maintainable automation components.
- **AIML Teaching Assistant — LNMIIT**
Taught and evaluated AI/ML laboratory sessions for 60 students
 - Taught and evaluated Python-based ML implementations spanning classification, regression, clustering, neural networks/CNNs, time series, search, and probabilistic reasoning.

PROJECTS

- **PaymentPulse — Payments Data & MLOps Platform** *Python, SQL, DuckDB, scikit-learn, MLflow, Docker, GitHub Actions, AWS*
 - **Data Pipeline:** Built an end-to-end data and ML pipeline processing 75K transactions through Bronze/Silver/Gold layers, implementing data curation, ETL, schema validation, deduplication, quarantine handling, and dimensional modeling.
 - **Data Quality & Controls:** Implemented 14 automated data-quality checks and 7 risk/control checks, achieving 100% DQ validation with zero reconciliation delta; automated regression testing through GitHub Actions and containerized execution with Docker.
 - **Model Operations:** Operationalized an Isolation Forest anomaly-detection workflow with behavioral feature engineering, model artifacts/metadata, automated validation gates, and PSI-based drift monitoring; designed an AWS deployment architecture using S3, SageMaker Pipelines, Lambda, Step Functions, and CloudWatch.
- **BookwiseAI — Retrieval-Augmented Generation Platform** *Python, FastAPI, LLMs, Vector Search, Docker*
 - **RAG Pipeline:** Built a document-grounded RAG application that ingests, indexes, retrieves, and generates responses from book content using semantic retrieval and LLM-based generation.
 - **API Deployment:** Developed a FastAPI backend and containerized the application with Docker, exposing an API-driven workflow for document retrieval and question answering.
- **Pokémon Blue AI Agent** *Python, Reinforcement Learning, Search, Game AI*
 - **AI Agent:** Developed an AI agent capable of interpreting game state and selecting actions through programmatic interaction with the game environment.
 - **Decision Logic:** Implemented search/planning and state-processing logic to automate decision-making across exploration and battle scenarios.

TECHNICAL SKILLS

- **Programming:** Python, SQL, Bash
- **ML / AI:** Machine Learning, Anomaly Detection, Computer Vision, CNNs, NLP, RAG, Model Evaluation
- **MLOps / DevOps:** MLflow, Model Validation, Model Monitoring, Drift Detection, Docker, GitHub Actions, CI/CD
- **Data / Cloud:** ETL/ELT, Data Pipelines, Data Quality, Data Governance, Data Lineage, AWS S3, SageMaker Pipelines, Lambda, Step Functions, CloudWatch
- **Databases / Frameworks:** PostgreSQL, SQL Server, DuckDB, MongoDB, FastAPI, pandas, scikit-learn, OpenCV

EDUCATION

- **The LNM Institute of Information Technology (LNMIIT)** Jaipur, India
Bachelor of Technology — Communication and Computer Engineering; 2023–2027
- **White Leaf Public School** India
CBSE — Class XII; 91.60%
- **Indus Valley Public School** India
CBSE — Class X; 95.30%

LEADERSHIP

- **Coordinator, Literary Committee** LNMIIT
Leadership, event operations, and stakeholder coordination
 - Led planning, scheduling, and cross-team communication for a 30+ member team, coordinating institute-wide events for 500+ participants.
- **Associate Coordinator, Counselling Cell** LNMIIT
Student outreach and peer-wellness initiatives
 - Supported student outreach and coordinated communication to promote awareness of available resources.